Conjecture 00000000742: minimal components and minimal maps on $\mathbf{Z}_2$

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1 Two readings of the statement

The conjecture gives p^{p-1} as the number of "minimal decompositions (minimal dynamical systems)" of 1-Lipschitz maps on $\mathbf{Z}_p$, and ends with "a count of minimal maps". These phrases can refer to different objects. At $p = 2$ the proposed number is 2. We refute both the count of minimal components of a fixed map and the count of distinct minimal maps. We do not identify different maps modulo topological conjugacy: the source specifies no such equivalence relation.

A minimal closed subsystem means a nonempty closed forward-invariant set K such that every forward orbit from a point of K is dense in K. In the terminology of Fan–Liao [1], their minimal components are certain clopen minimal subsystems; an everywhere minimal system has one component. Their general decomposition theorem concerns polynomials of degree at least two. We use their terminology only, and prove the affine examples below directly without invoking that theorem.

2 A single map has one minimal component

Let $T : \mathbf{Z}_2 \to \mathbf{Z}_2$ be $T(x) = x+1$. Translation preserves the 2-adic distance, so T is 1-Lipschitz. For every $n \geq 0$,

$$T^n(x) = x + n.$$

The nonnegative integers are dense in $\mathbf{Z}_2$: every residue class modulo 2^k has a nonnegative integer representative. Translation is a homeomorphism. It follows that every forward orbit $\{x + n : n \geq 0\}$ is dense in the actual 2-adic topology.

If K is any nonempty closed minimal subsystem, choose $x \in K$. Its orbit closure is both K and $\mathbf{Z}_2$, so $K = \mathbf{Z}_2$. Conversely, the whole space is a minimal subsystem, and is clopen. Thus the collection of nonempty closed minimal subsystems is exactly $\{\mathbf{Z}_2\}$. Its cardinality is 1, whereas $2^{2-1} = 2$. This also proves the assertion for the smaller class of clopen minimal components. No finite quotient is substituted for $\mathbf{Z}_2$.

3 There are at least three distinct minimal maps

For a unit $a \in \mathbf{Z}_2^\times$, let $T_a(x) = x+a$. This map is 1-Lipschitz and $T_a^n(x) = x+na$. Multiplication by a unit is a continuous surjection of $\mathbf{Z}_2$ (in fact a homeomorphism). Consequently it carries the dense nonnegative integers to a dense set, and translation by x shows that every forward T_a-orbit is dense.

The values $a = 1, 3, 5$ are all units. Thus T_1, T_3, T_5 are minimal 1-Lipschitz maps. Their values at zero are distinct in $\mathbf{Z}_2$ (which has characteristic zero), so the maps themselves are distinct. There are at least three such maps; there cannot be a bijection between the set of

these maps and a two-element set. These three examples are mutually conjugate, which is irrelevant to counting actual maps. We make no claim here about a count of conjugacy classes.

4 Formalization and verification

The Lean type `Z2` is Mathlib's actual `PadicInt 2`, with its 2-adic metric and topology. `odometer_lipschitz` proves the metric inequality, `odometer_iterate` proves the orbit formula, and `odometer_dense_orbit` uses the proved Mathlib theorem `PadicInt.denseRange_natCast`. The definition `IsMinimalComponent` states the closed-subsystem conditions above. `odometer_minimal_iff` characterizes every such set as the whole space; `minimal_component_count` proves its actual cardinality.

`IsMinimalMap` includes both the 1-Lipschitz property and the density of every forward orbit. `translation_dense_orbit` proves density for every unit translation. `threeMinimalMaps` constructs the three actual maps, and its injectivity is proved by evaluation at zero. `not_exactly_two_minimal_maps` rules out a bijection with the proposed finite set, without interpreting an infinite cardinality as a natural number.

All eleven principal declarations were audited. The package pins Lean 4.33.0, Mathlib and all transitive dependencies. A separate project build and official fresh-kernel replay accompany the source; no admitted proof, native decision shortcut or extra axiom is used. The bilingual source is preserved from upstream commit
`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

OpenAI Codex produced the proof and report. A separate supervising Codex agent reviewed both readings and the full formal chain. This is AI review, not human certification. The examples are classical and are not claimed as new mathematical results.

References

[1] A.-H. Fan and L. Liao, *On minimal decomposition of p-adic polynomial dynamical systems*, Advances in Mathematics 228 (2011), 2116–2144, Introduction. `https://arxiv.org/abs/1010.5583`.